



A note on series equivalent of the Riemann hypothesis

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Abstract In this manuscript we denote by $\sum_{\rho}$ a sum over the non trivial zeros of Riemann zeta function (or over the zeros of Riemann's xi function), where the zeros of multiplicity k are counted k times. We prove a result that the Riemann Hypothesis is true if and only if

$$\sum_{\rho} \frac{1}{|\frac{1}{2} - \rho|^2} = \frac{\xi''(\frac{1}{2})}{\xi(\frac{1}{2})}$$

Keywords Riemann zeta function · Riemann xi function · Riemann Hypothesis · Hadamard product

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1 Introduction

The Riemann zeta function, $\zeta(s)$ is defined as the analytic continuation of the Dirichlet series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

which converges in the half plane $\Re(s) > 1$. The Riemann zeta function is a meromorphic function on the whole complex s -plane, which is holomorphic everywhere except for a simple pole at $s = 1$ with residue 1. The Riemann Hypothesis states that all the non trivial zeros of the Riemann zeta function lie on the critical line $\Re(s) = \frac{1}{2}$. The Riemann xi function is defined as

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s)$$

$\xi(s)$ is an entire function whose zeros are the non trivial zeros of $\zeta(s)$ [2]. Further $\xi(s)$ satisfies the functional equation [2]

$$\xi(s) = \xi(1-s)$$

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2 Main result

The goal of this short note is to prove the following result.

Theorem If $\sum_{\rho}$ denotes a sum over the non trivial zeros of Riemann zeta function then the Riemann Hypothesis is true if and only if

$$\sum_{\rho} \frac{1}{|\frac{1}{2} - \rho|^2} = \frac{\xi''(\frac{1}{2})}{\xi(\frac{1}{2})}$$

Proof Riemann xi function $\xi(s)$ is defined as,

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) \quad (1)$$

Riemann xi function is also expressed as the Hadamard product [1]:

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right) \quad (2)$$

where ρ ranges over all the roots ρ of $\xi(\rho) = 0$. Taking logarithm in both sides of equation (2):

$$\log(\xi(s)) = \log(\xi(0)) + \sum_{\rho} \log\left(1 - \frac{s}{\rho}\right)$$

Considering s such that $\xi(s) \neq 0$ and differentiating both sides with respect to s :

$$\frac{\xi'(s)}{\xi(s)} = \sum_{\rho} \frac{1}{s - \rho}$$

Differentiating both sides again with respect to s :

$$\frac{\xi(s)\xi''(s) - (\xi'(s))^2}{(\xi(s))^2} = - \sum_{\rho} \frac{1}{(s - \rho)^2} \quad (3)$$

Lemma We have

$$\xi'\left(\frac{1}{2}\right) = 0$$

Proof Since the Riemann xi function, $\xi(s)$ satisfies the functional equation

$$\xi(s) = \xi(1-s)$$

We have since $\xi(s)$ is entire, so by differentiating the above equation with respect to s ,

$$\xi'(s) = -\xi'(1-s) \quad (4)$$

Now putting $s = \frac{1}{2}$ in equation (4), we have

$$\xi'\left(\frac{1}{2}\right) = -\xi'\left(\frac{1}{2}\right)$$

So we get,

$$\xi'\left(\frac{1}{2}\right) = 0$$

We have a more general result that the odd order derivatives of the Riemann xi function at $s = \frac{1}{2}$ vanishes [4, p. 528].

This completes the proof of Lemma.



Using $\xi'(\frac{1}{2}) = 0$ from Lemma we have in equation (3),

$$\frac{\xi''(\frac{1}{2})}{\xi(\frac{1}{2})} = - \sum_{\rho} \frac{1}{(\frac{1}{2} - \rho)^2} \quad (5)$$

Taking real parts of both sides of equation (5) and noting that $\frac{\xi''(\frac{1}{2})}{\xi(\frac{1}{2})}$ is real, we have:

$$\frac{\xi''(\frac{1}{2})}{\xi(\frac{1}{2})} = -\frac{1}{2} \left[\sum_{\rho} \frac{1}{(\frac{1}{2} - \rho)^2} + \sum_{\bar{\rho}} \frac{1}{(\frac{1}{2} - \bar{\rho})^2} \right] \quad (6)$$

Above equation gives

$$\begin{aligned} -2 \frac{\xi''(\frac{1}{2})}{\xi(\frac{1}{2})} &= \sum_{\rho} \frac{(1 - 2\Re(\rho))^2 - 2|\frac{1}{2} - \rho|^2}{|\frac{1}{2} - \rho|^4} \\ \sum_{\rho} \frac{(1 - 2\Re(\rho))^2}{|\frac{1}{2} - \rho|^4} &= 2 \sum_{\rho} \frac{1}{|\frac{1}{2} - \rho|^2} - 2 \frac{\xi''(\frac{1}{2})}{\xi(\frac{1}{2})} \end{aligned} \quad (7)$$

We prove that Riemann Hypothesis is true if and only if

$$\sum_{\rho} \frac{1}{|\frac{1}{2} - \rho|^2} = \frac{\xi''(\frac{1}{2})}{\xi(\frac{1}{2})} \quad (8)$$

Assume Riemann Hypothesis is true then we have $\Re(\rho) = \frac{1}{2}$ for all ρ and hence by equation (7) we get the result. Conversely suppose that equation (8) holds then by equation (7) we have

$$\sum_{\rho} \frac{(1 - 2\Re(\rho))^2}{|\frac{1}{2} - \rho|^4} = 0 \quad (9)$$

Write

$$T = \sum_{\rho} \frac{(1 - 2\Re(\rho))^2}{|\frac{1}{2} - \rho|^4}$$

Since the non trivial zeros of Riemann zeta function are countable, assume there exists atleast one ρ_k such that $1 - 2\Re(\rho_k) \neq 0$. Then for $n \geq k$, let the partial sum be T_n .

$$\text{So, } T_n \geq \frac{(1 - 2\Re(\rho_k))^2}{|\frac{1}{2} - \rho_k|^4} > 0$$

Thus

$$\lim_{n \rightarrow \infty} T_n = T > 0$$

which contradicts the fact that $T = 0$.

$$\text{So, } \Re(\rho_n) = \frac{1}{2} \text{ for all } n.$$

References

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